

Data Analytics and Visualization - Introduction



SoftUni Team
Technical Trainers



SoftUni



Software University

<https://about.softuni.bg>

Have Questions?

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#DAV

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A Day in My Work Life After This Course

- Who am I
- Who do I talk to?
- What problems do I solve?
- What topics do I care about?
- What should I do?



What skills a Data Analyst needs?

Time Management

Critical thinking

Collaboration

Storytelling

Communication

Attention to Details

Soft Skills

Hard Skills

Mathematics and Statistics

Data Manipulation & Analysis

Data Visualization

Working with Databases

Programing

Machine Learning*

Our Toolset

What we will use?

- We will use industry-standard tools.
- Database: PostgreSQL.
- SQL Client: DBeaver.
- BI Tool: Power BI Desktop.



Why PostgreSQL?

- Open Source & Free.
- Used by companies like Apple, Instagram, Spotify.
- Strict SQL compliance (Great for learning).



PostgreSQL

Why DBeaver?



- Universal Database Tool.
- Can connect to Postgres, MySQL, Oracle, etc.
- Clean Interface.



Why PowerBI?

- 
- User-Friendly Interface
 - Data Integration
 - Advanced Analytics
 - Customizable Visualizations
 - Cost-Effective



Power BI

Live Demo





What Data is ...
...and why is important

What Data is?

Definition: Raw facts and figures without context.

- **Numbers**

- 124,564.23
- 74%

- **Dates**

- 1/1/2025

- **Text Values**

- SoftUni
- Blue

- **Boolean Values**

- True / False



! Without context, data is useless.

- **Definition:** Highly organized, searchable, lives in rows and columns.
- **Examples:** SQL Databases, Excel, CSV.
- **Pros:** Easy to analyze, strict rules.

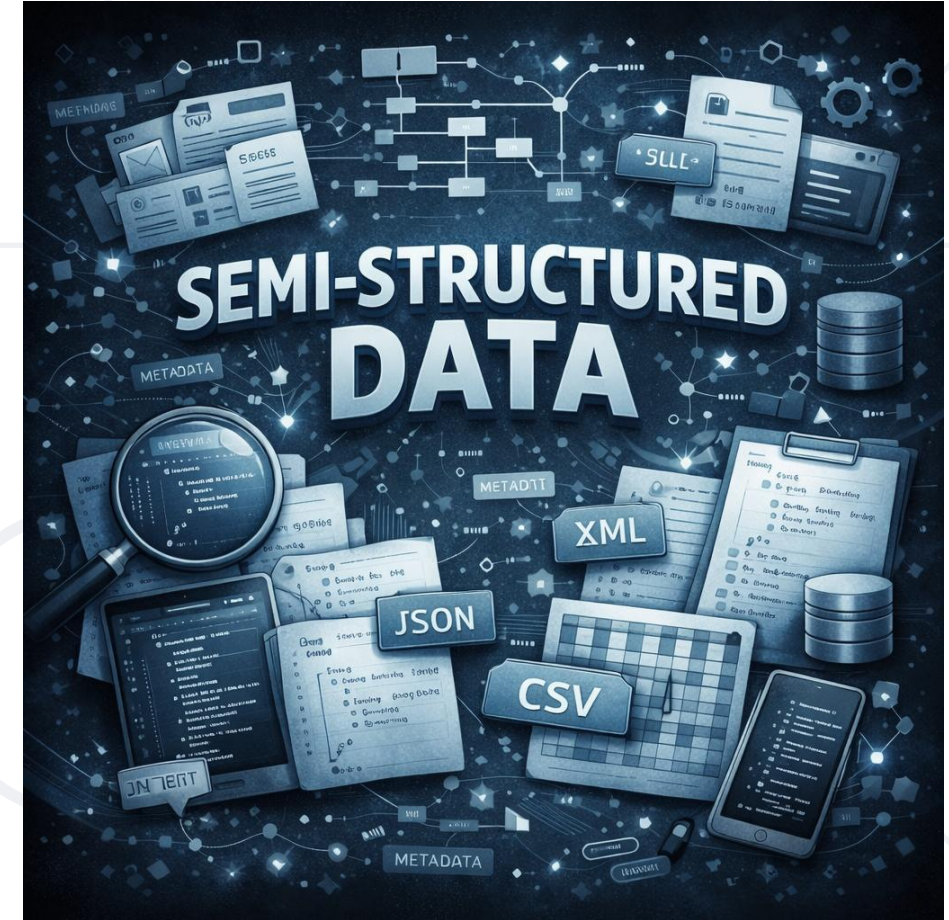
ID (Number)	Date (YYYY-MM-DD)	Customer Name (Text)	Product (Text)	Amount (Currency)
1001	2023-10-25	Ivan Ivanov	Laptop X	\$1250.00
1002	2023-10-25	Ivan Ivanov	Laptop X	\$1250.00
1003	2023-10-25	Ivan Ivanov	Laptop X	\$1250.00
1004	2023-10-25	Ivan Ivanov	Laptop X	\$1250.00
1005	2023-10-25	Ivan Ivanov	Laptop X	\$1250.00
1006	2023-10-25	Ivan Ivanov	Laptop Y	\$1250.00
1007	2023-10-25	Ivan Ivanov	Laptop X	\$1250.00

- **Definition:** No predefined format. Messy.
- **Examples:** Emails, Videos, PDF Documents, Social Media posts.
- **Challenge:** Hard to analyze directly with BI tools.

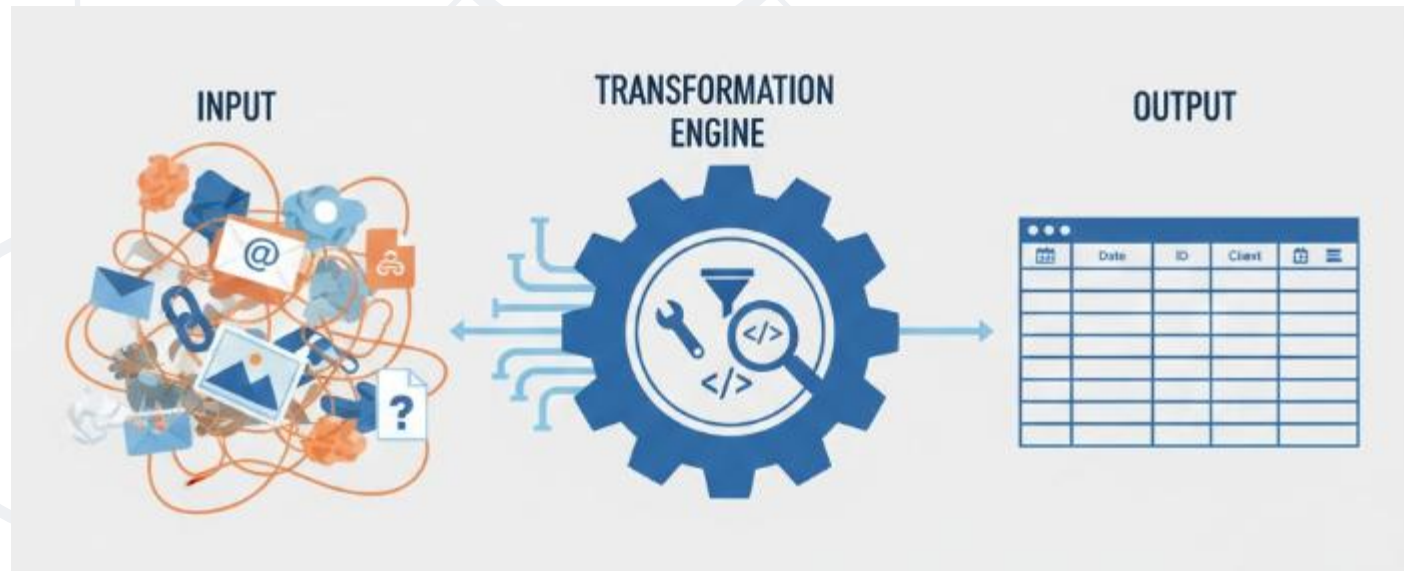


Semi-Structured Data

- **Definition:** Contains tags/markers but no strict table structure.
- **Examples:** JSON, XML.
- **Context:** Used heavily in Web APIs (e.g., getting weather data).



- Power BI loves **Structured Data**.
- **Our Job:** We are engineers. We turn Unstructured/Messy data -
> Structured Data.



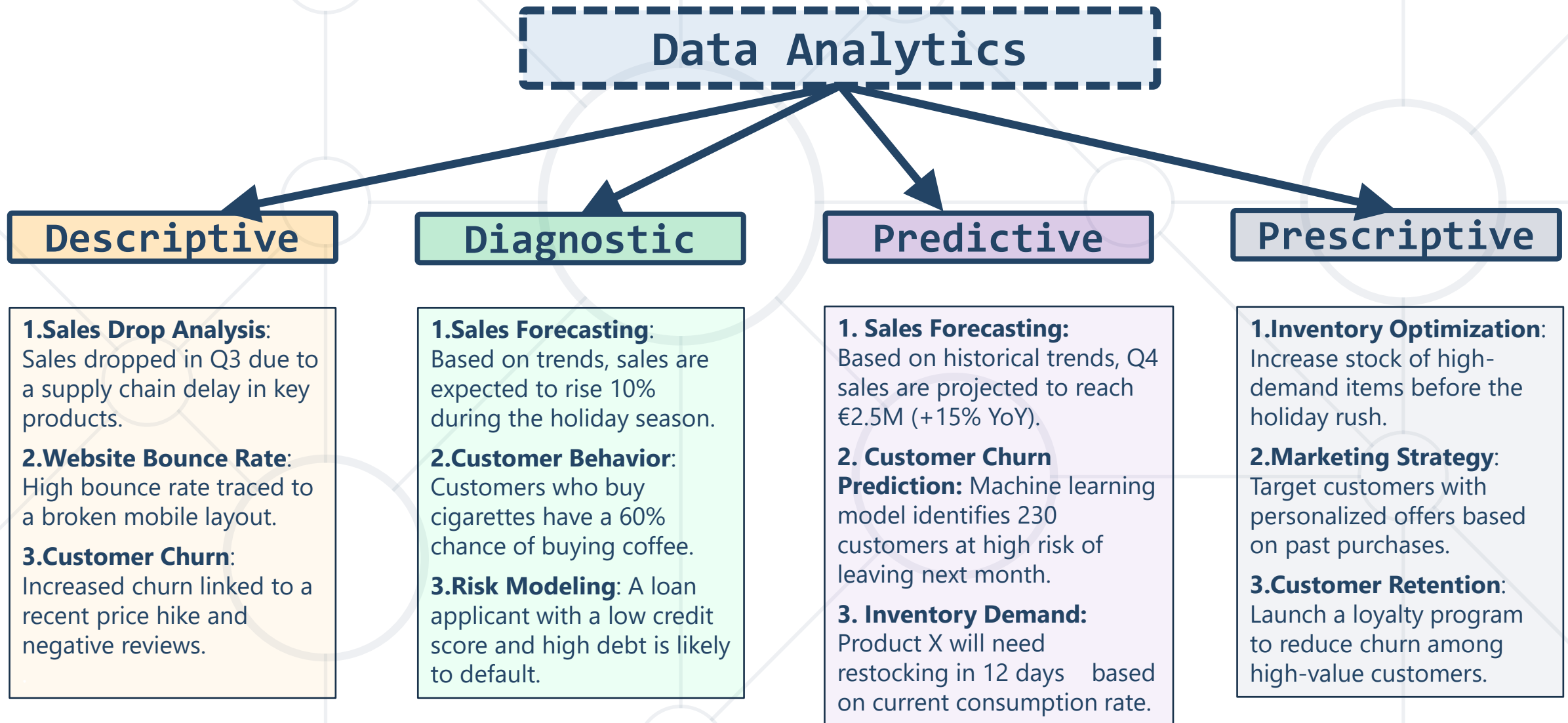
Live Demo



Why Do We Analyze Data?

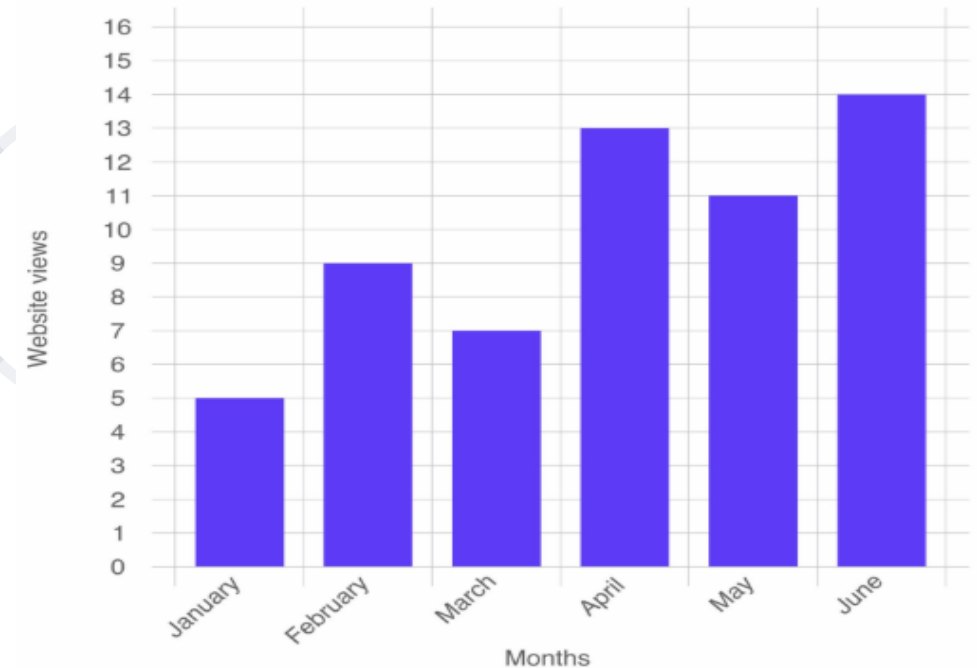
- To solve business problems.
- To move from Gut Feeling to Data-Driven Decisions.





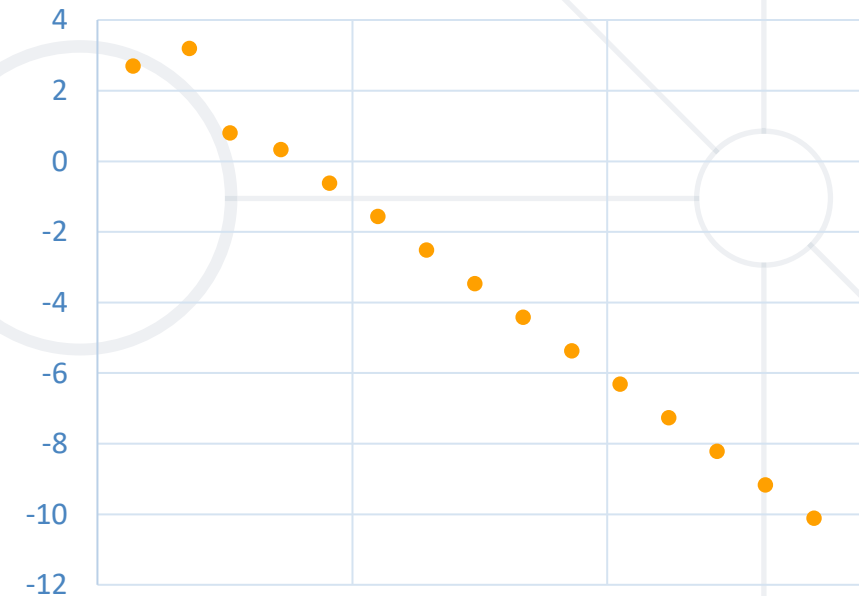
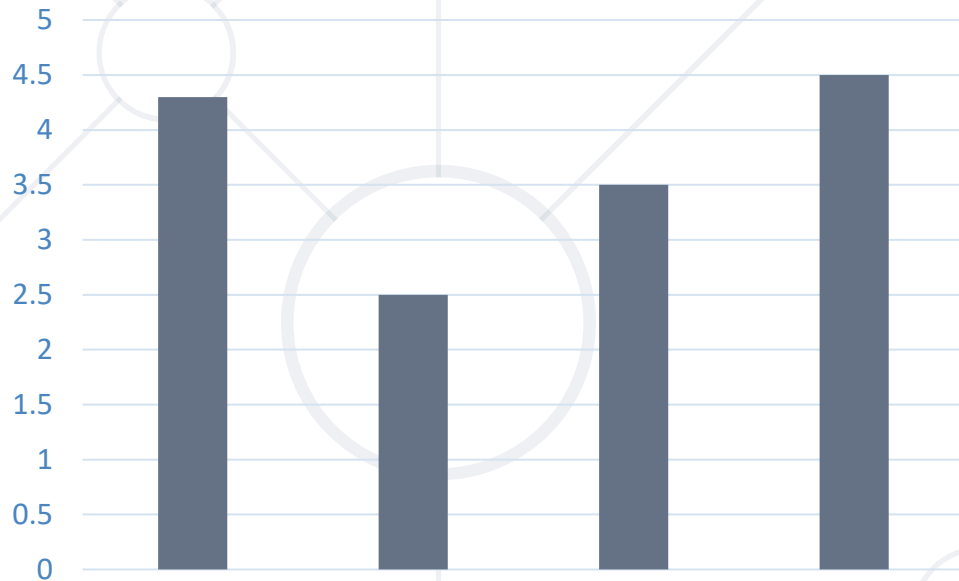
Level 1 - Descriptive Analytics

- Question: What happened?
- Output: Reports, Dashboards.
- Example: "Sales dropped by 10% last month."



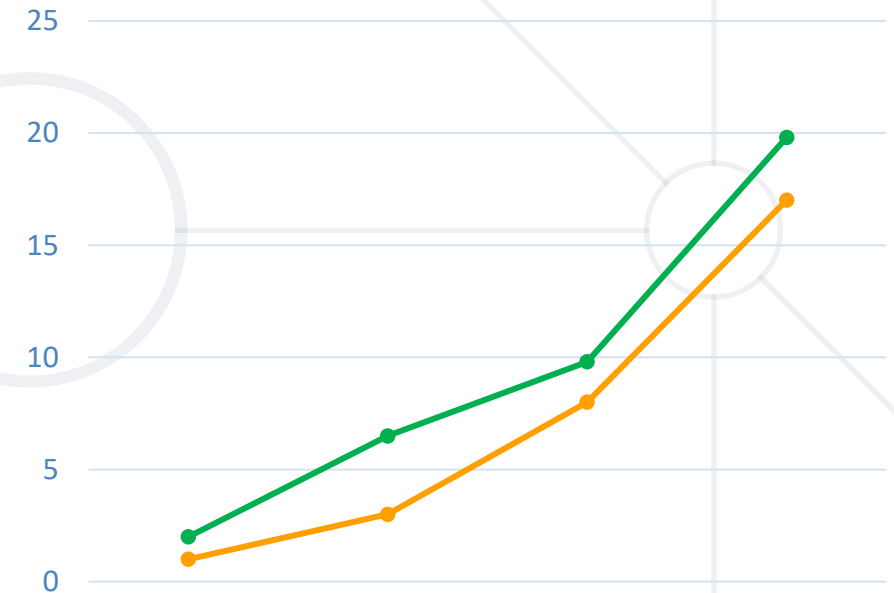
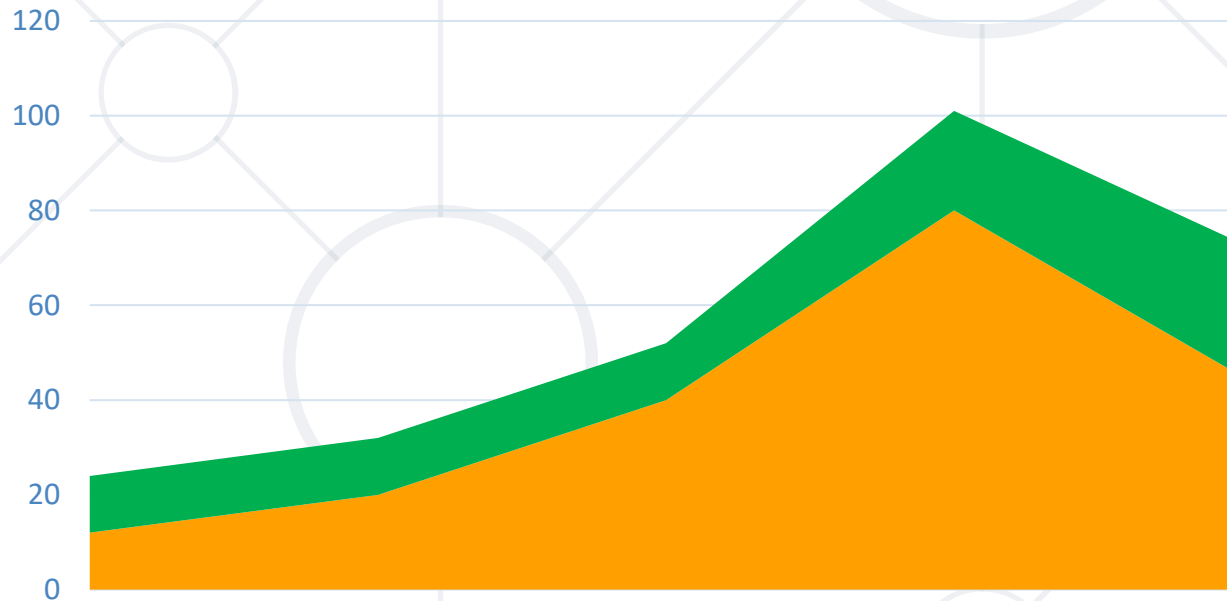
Level 2 - Diagnostic Analytics

- Question: Why did it happen?
- Output: Drill-down, Correlations.
- Example: "Sales dropped because the 'Sofia' store was closed for renovation."



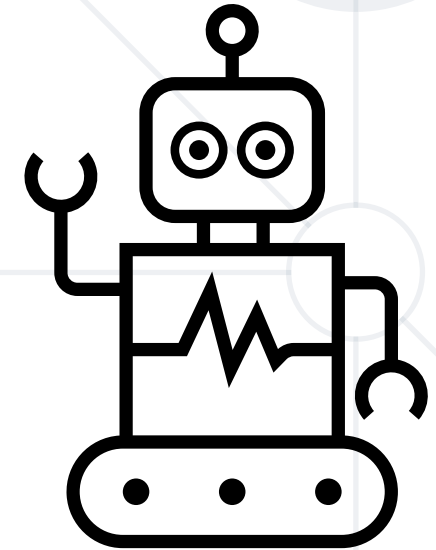
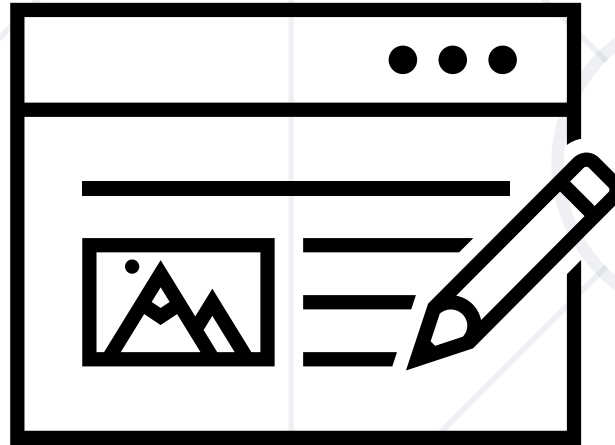
Level 3 - Predictive Analytics

- Question: What will happen?
- Output: Forecasts, Trends.
- Example: "Sales will likely recover in Q4."



Level 4 - Prescriptive Analytics

- Question: How can we make it happen?
- Output: Recommendations, Automation.
- Example: "Automatically order more stock for the Sofia store."



Where does Power BI fit?

- Power BI is excellent for Descriptive and Diagnostic.
- It has some Predictive features (Forecasting lines).



Live Demo



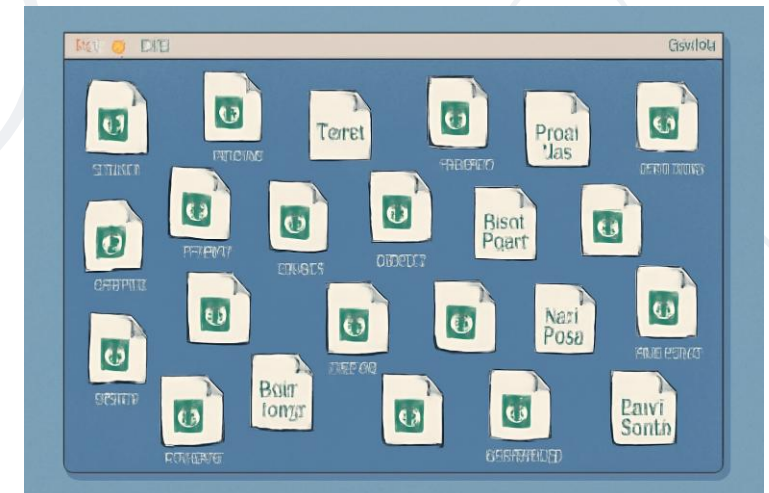


The Modern Data Stack

The "Old Way" (Problem)

The Reality in Many Companies:

- **Manual Work:** Copy-pasting data from CRM to Excel daily.
- **Silos:** Marketing has one file, Sales has another. Numbers don't match.
- **Fragile:** One broken formula destroys the report.
- **Not Scalable:** Excel crashes after 1 million rows.





- Scenario: Marketing has one Excel file. Sales has another.
- Result: "Why don't the numbers match?,"
- Issues: Security risks, manual copy-paste, crashes on large data.

The "New Way" (Solution)

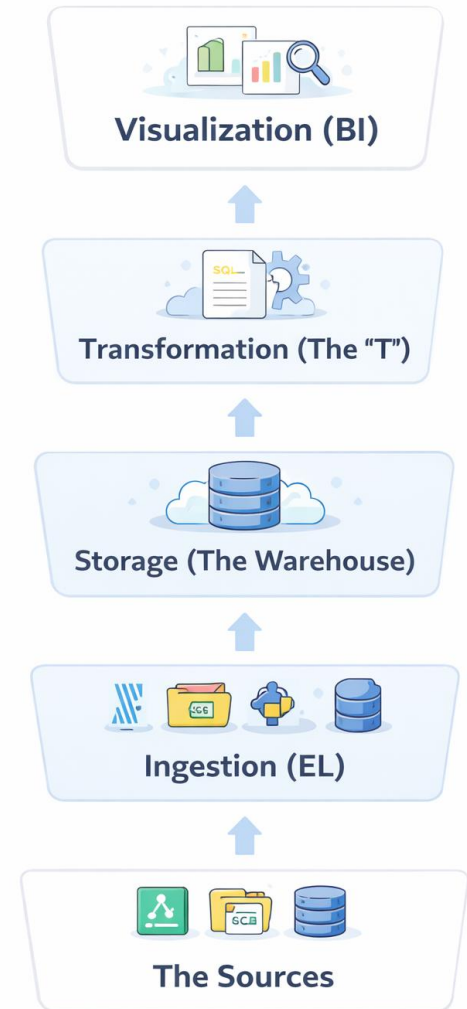
The Modern Data Stack (MDS)

Definition: A set of tools that work together to create a data platform.

Goal: Single Source of Truth.

Principles of Engineering:

- **Automated:** No human touch needed to move data.
- **Centralized:** One "Single Source of Truth" (Data Warehouse).
- **Scalable:** Can handle Petabytes of data.
- **Reproducible:** Logic is code (SQL), not clicks.



Layer 1: The Sources

- Where data is born.
- Apps: Salesforce, ERPs.
- Files: Excel, CSV.
- Databases: Production DBs (Postgres/MySQL).



The Sources

Layer 2: Ingestion (EL)

- Definition: Moving data from Source to Storage.
- Tools: Fivetran, Airbyte, Python Scripts.
- Principle: "Don't touch it, just move it."



Layer 3: Storage (The Warehouse)

- Definition: Centralized storage for analytics.
- Tools: PostgreSQL, Snowflake, BigQuery.
- Why Postgres? Powerful, Open Source, Standard SQL.



Storage (The Warehouse)

Layer 4: Transformation (The "T")

- Definition: Cleaning and modeling the data using SQL.
- Tools: dbt, SQL Views, Stored Procedures.
- This is where "Raw Data" becomes "Gold Data".



Transformation (The "T")

Layer 5: Visualization (BI)

- Definition: Presenting data to the end-user.
- Tools: Power BI, Tableau, Looker.
- The tip of the iceberg.



Visualization (BI)

ETL vs. ELT (Old vs New)

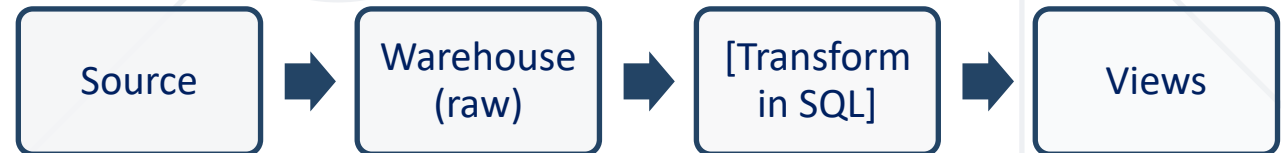
ETL (Extract-Transform-Load): Old school. Transform before loading.

- Pros: Saves disk space.
- Cons: Slow, inflexible.



ELT (Extract-Load-Transform): Modern. Load everything raw, transform later in SQL.

- Pros: Fast, agile, audit trail.



Live Demo

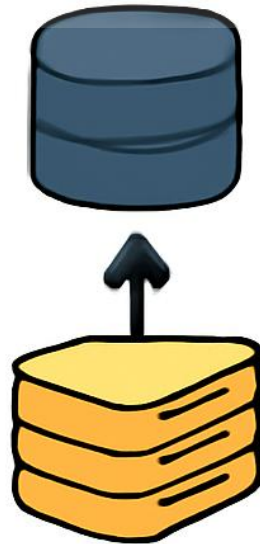


Connecting Power BI

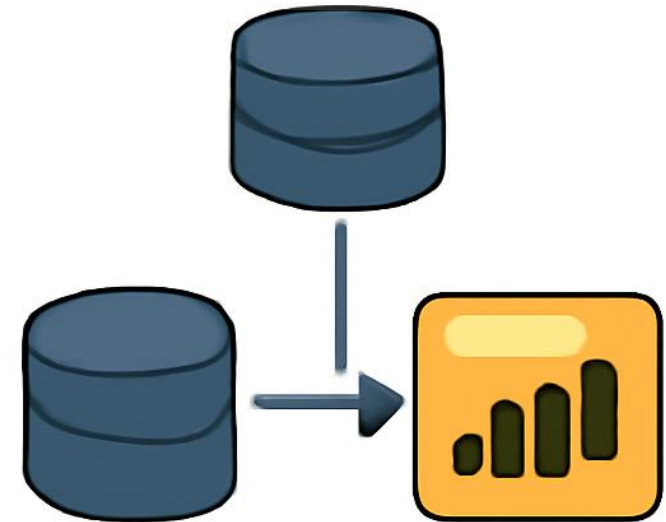
Once we have a database (PostgreSQL), how does Power BI read it?

Two Main Approaches:

Import Mode
(The Cache)

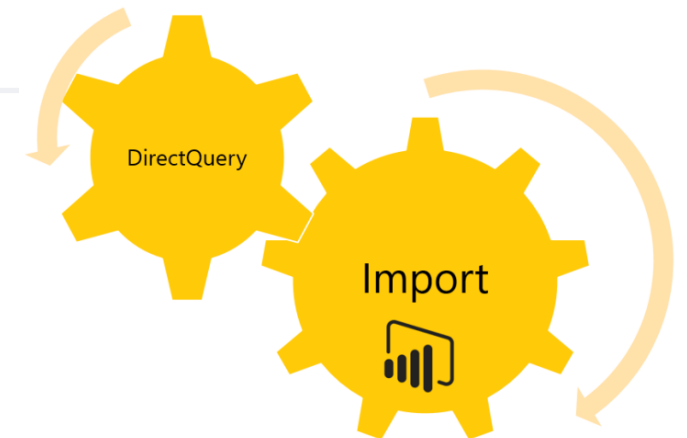


DirectQuery
(The Live Link)



Import Mode

- Definition: Power BI takes a snapshot of the data and stores it in its own memory (RAM).
- Analogy: Taking a screenshot of a website. It doesn't change if the website changes.
- The connection is closed after loading.





- Speed: In-memory calculation is blazing fast.
- Functionality: Supports all DAX functions.
- Modeling: You can combine multiple sources easily.

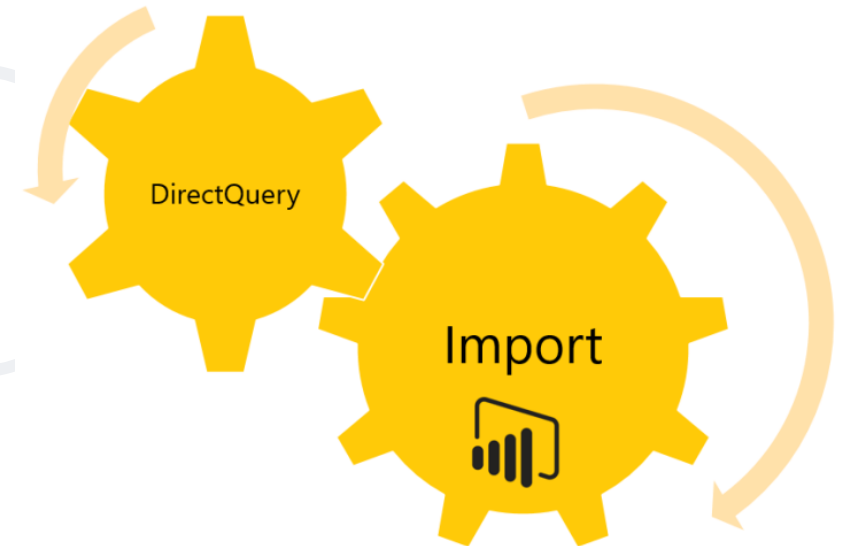


- Data Freshness: Not real-time (needs scheduled refresh).
- Size Limit: Depends on your RAM / License capacity.



DirectQuery Mode

- Definition: Power BI stores NO data. It sends SQL queries to PostgreSQL every time you click a visual.
- Analogy: Browsing a live website. You see changes instantly.





- **Real-time:** Data is always up to date.
- **Big Data:** Can handle Petabytes of data (since it stays in the DB) without downloading it.



- **Performance:** Slower. Depends on how fast the Database is.
- **Limitations:** Some DAX functions don't work.



Comparison Table

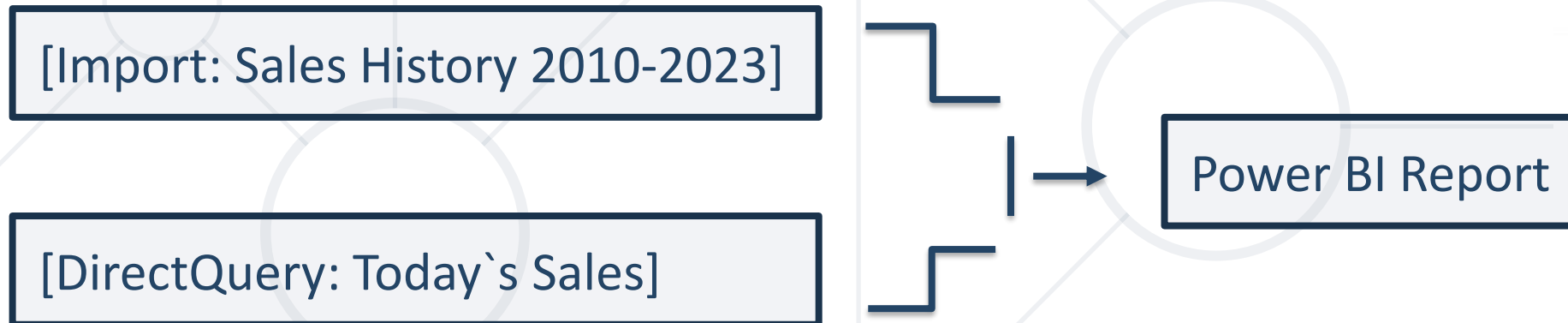


Feature	Import Mode	DirectQuery
Speed	⚡ Very Fast	🐢 Slower (Network dependent)
Data Size	Limited (RAM)	Unlimited (Source)
Freshness	Scheduled Refresh	Real-time
Use Case	90% of projects	Banks, Real-time monitoring

Composite Models (Hybrid)

The Best of Both Worlds.

- Keep "Sales History" (2010-2023) in Import.
- Keep "Today's Sales" in DirectQuery.



- When to use what?
- Under 1 GB? -> Always Import.
- Real-time needed? -> DirectQuery.
- Slow Database? -> Import.

Scenario	Import	DirectQuery
< 1 GB data	✓	✗
Real-time needed	✗	✓
Slow database	✓	✗
Complex DAX needed	✓	✗
Petabyte scale	✗	✓

Live Demo



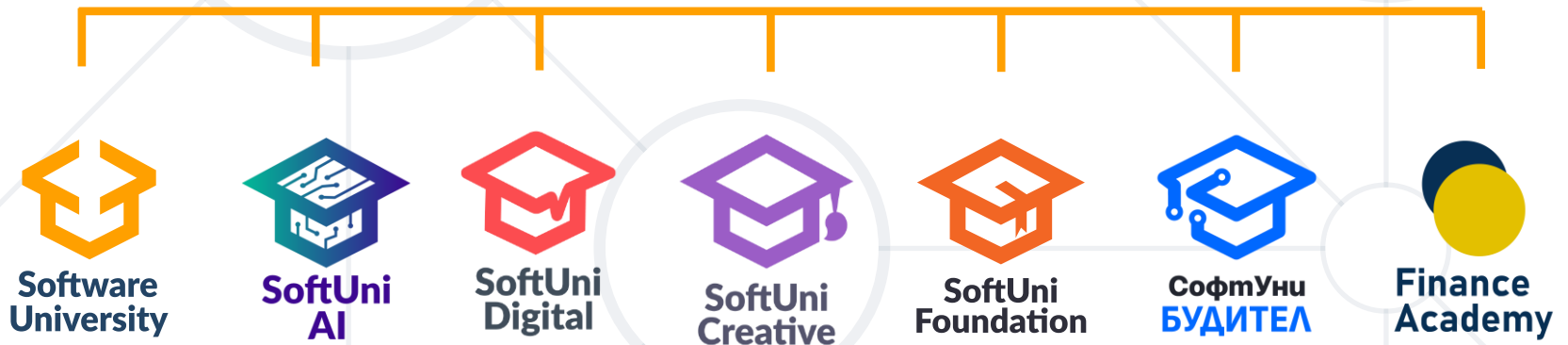
- Data must be Structured.
- We use the ELT pattern (Load raw, Transform in SQL).
- Import Mode is usually the default choice for Power BI.
- PostgreSQL is our backend engine.



Questions?



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